

Independent Study/ Reading Course Registration Form

Course No: CS4002

Course Title: physio-placebo: Is the LLM Reading the Signal, or Is the Verbalizer Doing the Classifying?

Instructor: Dr. Siddharth

Student Name: Shriyak Kumar

ID No: _____

Major: CSAI

- 1. Course Description:** Please include a list of topics, learning outcomes, and list of reference materials. (Include any additional information as attachments)

This independent study is a surrogate-controlled audit of the fast-growing "LLMs reason over your physiology" literature. Published prompting pipelines (Health-LLM style) are run under strict leave-one-subject-out (LOSO) evaluation on three public wearable datasets, and the physiological signal is then systematically replaced with spectrum-matched noise, another subject's recording, a flatline, or nothing at all, to measure how much of the above-chance performance survives. The central contribution is separating **verbalizer leakage** (class-suggestive adjectives introduced by feature-to-text templates, e.g. "elevated skin conductance, stress-typical") from genuine signal information. The study culminates in a publishable audit paper and an open-source evaluation package (physio-placebo).

Topics Covered:

- Physiological signal processing for wearables: ECG, PPG/BVP, EDA, and respiration; HRV time- and frequency-domain features; tonic/phasic EDA decomposition (NeuroKit2, cvxEDA)
- Affective and cognitive state classification: stress (WESAD, UCI), mental workload (MAUS, IEEE DataPort), cognitive effort (CogWear, PhysioNet) under subject-disjoint LOSO protocols
- LLM evaluation methodology: prompting paradigms (raw numeric series, verbalized features, plot images), few-shot prompt selection, deterministic logprob-based label scoring
- Surrogate-control and null-model methodology: moment-matched Gaussian noise, IAAFT phase randomization, subject permutation with broken label alignment, flatline, and signal deletion
- Verbalizer-leakage analysis: adjective neutralization and adversarially mis-signed feature templates
- Classical baselines: regularized logistic regression and shallow decision trees on identical features
- Statistics for small-subject-count studies: macro-F1, subject-level bootstrap confidence intervals, Wilcoxon signed-rank tests, permutation-based chance floors
- Open-science practice: OSF pre-registration, reproducible pipelines, open-source release; (stretch) parameter-efficient fine-tuning with QLoRA

Learning Outcomes: By the end of this independent study, the student will:

- Build a reproducible, leakage-free LOSO evaluation harness for physiological state classification
- Implement and validate surrogate (null) signal conditions, and quantify how much class information each surrogate retains
- Reproduce a published Health-LLM headline result and report the reproduction gap honestly
- Design, pre-register (OSF), and execute a falsifiable experimental protocol with frozen analysis decisions
- Apply subject-level uncertainty quantification appropriate for datasets of 11-22 subjects
- Write a publication-quality audit paper and release a pip-installable evaluation package

Reference Materials:

- Jo et al., "Are EEG-to-Text Models Working?" (arXiv:2405.06459), the noise-baseline audit discipline this project imports
- Tan et al., "Are Language Models Actually Useful for Time Series Forecasting?" (NeurIPS 2024)
- The Health-LLM line of prompting pipelines over wearable features (exact reproduction target pinned in Week 1 from the released code)
- SensorLM: sensor-language foundation models (arXiv:2506.09108, NeurIPS 2025), including its released captioning templates
- Datasets: WESAD (UCI ML Repository), MAUS (IEEE DataPort), CogWear (PhysioNet)
- Tooling: NeuroKit2, cvxEDA, vLLM, Unsloth; models: Qwen3-8B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-VL-7B-Instruct

2. Schedule and Course Plan: Students must meet with the Instructor at least once a week.

The student will meet with the instructor at least once a week to review progress, resolve design decisions, and receive feedback. Two hard gates structure the semester: the OSF pre-registration at the end of Week 3 (no surrogate condition is run before it), and the Week-10 minimum viable result (a complete, self-contained audit report).

Weekly Plan (15 weeks):

- **Week 1:** Download and validate all three datasets; freeze version-pinned feature extraction (HRV, EDA tonic/phasic, respiration); build the subject-disjoint LOSO harness; select the Health-LLM reproduction target.
- **Week 2:** Classical floor, locked before any LLM runs: logistic regression and depth-3 decision tree on identical features, with majority-class and permutation-based chance levels.
- **Week 3:** Fidelity anchor: reproduce one published Health-LLM headline number and report the gap; file the OSF pre-registration (metrics, surrogate definitions, validity gate, predictions).
- **Weeks 4-5:** Paradigm A (raw numeric series) and Paradigm B (verbalized features) with real signal, 2 open models x 3 datasets; prompt templates selected on a held-out fold, then frozen.
- **Week 6:** Paradigm C (signal plots read by a vision-language model) on WESAD and MAUS.
- **Weeks 7-8:** Full surrogate ladder across all paradigms, models, and datasets; information-preservation ledger (classical baseline run on every surrogate).
- **Week 9:** Verbalizer-leakage ablation: original vs neutralized vs adversarially mis-signed templates; item-level flip analysis.
- **Week 10:** Minimum viable result: the complete SDS report card (3 paradigms x 2 models x 3 datasets x 6 signal conditions) with full statistics; mid-semester check-in with the instructor.
- **Weeks 11-12:** Cross-dataset generalization; stretch datasets only if the pipeline is clean.
- **Week 13:** QLoRA fine-tuned condition (does fine-tuning make the model start using the signal?); optional GPT-class API upper bound.
- **Week 14:** Paper draft with all tables generated from committed scripts (targets: ACII/ICMI, or an ML4H/GenAI4Health workshop).
- **Week 15:** Open-source release of the physio-placebo package and reporting checklist; final report; reproduction script shared with the original authors.

3. Assessment Schedule and Grading:

- **Reproduction and Pre-Registration Package (20%)** - due end of Week 3: working data/feature/LOSO pipeline, locked classical baselines and chance floors, reproduction of one published Health-LLM number with the gap reported, and the filed OSF pre-registration.
- **Mid-Semester Audit Report, the SDS Report Card (30%)** - due end of Week 10: the full results grid (3 paradigms x 2 models x 3 datasets x 6 signal conditions) with subject-level bootstrap confidence intervals, Wilcoxon tests, classical-baseline deltas, and the verbalizer-ablation panel, presented at a check-in meeting with the instructor.
- **Final Paper (30%)** - due Week 15: a workshop-quality audit paper with an explicit limitations section, with every number generated from committed scripts.

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- **Open-Source Release and Reproducibility (10%)** - due Week 15: pip-installable physio-placebo package with a quickstart verified on a fresh machine, plus the reporting checklist.
- **Weekly Engagement (10%)** - sustained weekly meetings across the semester with evidence-backed progress updates.

4. Additional Information

- All datasets are public and de-identified (UCI ML Repository, IEEE DataPort, PhysioNet); the study involves no human-subjects data collection.
- The experimental design is pre-registered on OSF in Week 3; frozen components (surrogate definitions, metrics, validity gate, predictions) are not modified after registration.
- Compute: a single GPU with at least 16 GB VRAM; all LLM conditions are scored via logprobs over label tokens (approx. 20-40 GPU-hours for the full grid); optional commercial-API budget under \$50.
- The paper follows the target venue's format (ACII/ICMI, or ML4H/GenAI4Health workshop); all code and results are version-controlled with a fully reproducible pipeline.
- Weekly meetings are held with notes recorded for progress tracking; a structured check-in template is used for each meeting.

5. Category: Program Elective

By signing below, I confirm that I have reviewed the course requirements, the assessment schedule, and understand the commitment involved.

Student Signature: _Shriyak Kumar_

Date: _10/08/2026_

For Office Use Only:

Approved by: Choose an item.

Signature: _____

Date: _____

Approved by Office of Academic Affairs: ☐